

Leveraged ETF Momentum Strategy

Quantitative Portfolio Management

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Abstract

Systematic strategy for a leveraged ETF sleeve of TECL (3x Technology), ROM (2x Technology), and NAIL (3x Homebuilders), equal-weighted. Position is scaled from zero to full exposure by two gates and one graduated overlay: (1) a trend gate — reference index above its 200-day simple moving average, (2) a volatility gate — 20-day realised volatility below 22%, and (3) a three-state VVIX overlay — RISK_UP / RISK_MID / RISK_OFF at 100 and 160 — that scales exposure to 100%, 50%, or 0% (full exit at extreme VVIX) when the gates are open. Risk-off cash sits in SGOV. Backtested on 10.9 years: Sharpe 1.34, CAGR 31%, MaxDD -27% after 5 basis points of transaction cost per rebalance. Live sizing capped at 10% of net asset value under a Kelly-quarter fractional cap.

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1 Executive Summary

Metric (net of 5 bps per trade)	Value
CAGR	30.7%
Sharpe ratio	1.34
Maximum drawdown	-26.7%
Volatility (annualised)	22.9%
Time at full exposure (RISK_UP)	44%
Time at reduced exposure (RISK_MID)	1%
Time in cash (regime OFF or VVIX ≥ 160)	55%
Rebalances per year	~9 (portfolio-level)
Total transaction cost over 10.9 years	2.4 percentage points
Stress correlation vs broad market (SPY)	0.20
Backtest window	2015-08 — 2026-07 (10.9 years)

Table 1: Recommended portfolio: TECL + ROM + NAIL equal-weight 33/33/33, with SMA + vol trend / regime gates and the 3-state VVIX overlay. Net of trading costs. Cross-strategy stress correlation of 0.20 confirms the sleeve decouples from broad-market drawdowns.

Pre-registered criteria — results:

Q	Question	Gate	Observed
A	Do sample-window results generalise?	$Sh \geq 0.80$	1.34 (net of TC)
B	Are the vol thresholds robust or overfit?	$OOS\ drop \leq 0.20$	static beats walk-fwd
C	Can drawdown be reduced without hurting CAGR?	$\geq 5pp\ DD\ gain$	VVIX 3-state: 8pp gain
D	Independent of the broad market under stress?	$Stress\ \rho \leq 0.60$	0.20
E	Recommended live sizing?	—	10% NAV, Kelly-quarter

Table 2: All four testable criteria pass.

Live sizing: 10% of net asset value, split equally across TECL, ROM, and NAIL. Kelly-quarter fractional, capped at 10% NAV. Monthly rebalance. Weekly signal check with Telegram alerts on state transitions. Manual execution during a paper trading period of at least 2 months (see ADR-0002).

2 Strategy Design

2.1 Why leveraged ETFs need active exposure management

Leveraged ETFs amplify daily index returns by 2x or 3x. Two structural risks:

- **Volatility decay:** daily rebalancing erodes value in choppy markets. A 3x ETF on an index that moves -10% then +10% loses more than the index due to path-dependent compounding.
- **Catastrophic drawdowns:** a 30% index decline maps to 60–80% loss in a 3x fund, requiring 150–400% to recover.

Systematic scaling of exposure — from zero to full — solves both. The strategy uses three signals in a fixed hierarchy: two binary gates that decide whether to hold any exposure at all, and one graduated overlay that decides how much.

2.2 The trend gate — 200-day simple moving average

A **simple moving average** (SMA) at window N is the arithmetic mean of the last N daily closing prices:

$$\text{SMA}_N(t) = \frac{1}{N} \sum_{i=0}^{N-1} P_{t-i}$$

The 200-day SMA is a widely-used long-term trend estimator that smooths out short-term noise. When the current price is above the 200-day SMA, the recent 40 weeks of trading have been net upward — a trending regime. When it is below, the recent history has been net downward — a broken trend.

The trend gate:

$$\text{Trend gate} = \begin{cases} \text{OPEN} & \text{if } P_t^{\text{ref}} > \text{SMA}_{200}(P^{\text{ref}}) \\ \text{CLOSED} & \text{otherwise} \end{cases}$$

where P^{ref} is the reference index (XLK for TECL and ROM, ITB for NAIL). We use the reference index, not the leveraged ETF itself, because a leveraged ETF's own SMA is distorted by daily-decay effects.

2.3 The volatility gate — 20-day realised vol

The 20-day annualised realised volatility of the reference index is:

$$\sigma^{(20)}(t) = \sqrt{252} \cdot \text{std}(\{r_{t-19}, \dots, r_t\})$$

where r_i is the daily log return. It measures how choppy the last 4 weeks have been.

The volatility gate:

$$\text{Vol gate} = \begin{cases} \text{OPEN} & \text{if } \sigma^{(20)}(P^{\text{ref}}) < \theta \\ \text{CLOSED} & \text{otherwise} \end{cases}$$

with $\theta = 22\%$ for both XLK and ITB. This threshold was validated by walk-forward analysis (Section 3).

Combined gate. Exposure is only allowed when *both* gates are open:

$$\text{Regime} = \begin{cases} \text{ON} & \text{if Trend gate} = \text{Vol gate} = \text{OPEN} \\ \text{OFF} & \text{otherwise} \end{cases}$$

2.4 The graduated overlay — three VVIX-driven exposure states

When both gates are open, the exposure level is set by VVIX (the CBOE VIX-of-VIX index — the implied volatility of 30-day VIX options). VVIX is a *leading* stress indicator that spikes days-to-weeks before realised volatility crosses the 22% threshold.

Three states:

State	VVIX close	Sleeve exposure
RISK_UP	< 100	100% of sleeve cap
RISK_MID	100 to 160	50% of sleeve cap
RISK_OFF-VVIX	≥ 160	0% (full exit)

Table 3: Three-state VVIX overlay. States are evaluated at Friday close; transitions apply on the following Monday. At extreme VVIX (≥ 160 , roughly the 99th percentile over 10.9 years) the option market is pricing catastrophic uncertainty and the sleeve exits to cash regardless of the trend / vol gates.

The final exposure combines all three signals as a product:

$$\text{Exposure}_t = \underbrace{\mathbb{1}_{\text{Trend gate OPEN}}}_{0 \text{ or } 1} \cdot \underbrace{\mathbb{1}_{\text{Vol gate OPEN}}}_{0 \text{ or } 1} \cdot \underbrace{s_{\text{VVIX}}(V_t)}_{\{1.0, 0.5, 0\}} \cdot w$$

where s_{VVIX} is the VVIX scale factor (1.0, 0.5, or 0) and w is the sleeve NAV cap (10%). The sleeve exits fully to cash whenever *any* of the three signals says exit: trend broken, or vol high, or VVIX in extreme panic. Otherwise exposure is 100% or 50% of the cap depending on VVIX bucket.

Why three states and not two or four. Binary halving (RISK_UP or RISK_MID only) gives Sharpe 1.34 but has no explicit exit response to extreme VVIX — it holds 50% exposure through a March 2020 event. That is not defensible. Adding RISK_OFF-VVIX at 160 restores the intuition (extreme uncertainty $\rightarrow$ exit) at zero cost to Sharpe (still 1.34). Four or more states adds trigger frequency without adding information — VVIX’s distribution has three natural modes (calm, elevated, panic), not five.

Why 160 not 130. At the 130 level, VVIX drops back below within a week or two roughly 80% of the time — most 130 events do not resolve into a full crash. Exiting at 130 costs 0.7 pp of CAGR for no material MaxDD improvement. At 160, only 18 days in 10.9 years qualify, and every one of them was during a genuine tail event (2015 China, 2018 volmageddon, 2020 COVID, 2022 rate cycle, 2025 tariff shock). Exit here is unambiguous.

2.5 Risk-off cash routing — SGOV

When Regime = OFF, sleeve cash is placed in SGOV (iShares 0–3 Month Treasury Bond ETF) to earn T-bill yield (currently $\sim 4\%$). Over the extended window, the strategy is in cash 55% of the time; SGOV routing contributes ~ 2 percentage points to CAGR at zero drawdown risk.

2.6 Rebalancing cadence

Signals evaluated on Friday close. Trades placed on the following Monday morning. Rebalances occur when:

- Regime state changes (ON $\leftrightarrow$ OFF)
- VVIX state changes (UP $\leftrightarrow$ MID, or entry into RISK_OFF at VVIX ≥ 160)
- First Friday of a new calendar month (rebalance to 33/33/33)
- Drift band: any leg $> \pm 5$ pp from its target weight

Trigger events per year (portfolio-level, over the extended backtest):

Event type	Frequency (per year)
Regime state changes (ON $\leftrightarrow$ OFF)	8–10
VVIX state changes (UP $\leftrightarrow$ MID)	1–2
VVIX RISK_OFF triggers ($VVIX \geq 160$)	0–1
Monthly rebalance	12 (calendar)
Drift-band rebalance	1–2
Total meaningful rebalances	~11 per year

Table 4: Rebalance frequency estimate. Roughly one meaningful action per month, plus the calendar rebalance.

3 Walk-Forward Validation

3.1 What walk-forward validation is

The 22% vol threshold is a hand-picked number. To check whether it was chosen because it happened to work on a particular window, we test it out-of-sample.

The procedure:

1. Split the data into rolling **3-year training** + **1-year test** windows, stepping forward by 1 year.
2. On each training window, grid-search the vol threshold over $\{0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.35\}$ and pick the value with the best training-window Sharpe.
3. Apply that trained threshold to the next 1-year test window — data the algorithm has not seen.
4. Roll forward. Chain the test-window returns into one out-of-sample track record.
5. Compare against a static 22% baseline.

If 22% was overfit, letting the algorithm re-pick would beat it. If it is a real regime cutoff, static should hold.

3.2 Result

Ticker	Static 0.22	Walk-forward	Adaptive 75%ile
TECL	0.94	0.76	0.63
NAIL	0.58	0.52	0.35
ROM	0.93	0.68	0.62
Mean (tradeable)	0.82	0.65	0.53

Table 5: Out-of-sample Sharpe by ticker, 2013–2026. Static 0.22 beats both alternatives on the core tradeable tickers. On TECL, the walk-forward chose 0.22 in 8 of 14 training windows on its own — the data keeps re-picking the same value.

Static 22% wins by 0.17 Sharpe on average $\rightarrow$ PASS. Adopted as the vol gate threshold for both XLK and ITB.

4 Backtest Results

4.1 Per-ticker (extended window)

Ticker	CAGR	Sharpe	MaxDD	Years	Status
TECL	32.2%	0.87	-45.5%	16.5	tradeable (core)
ROM	23.1%	0.84	-32.8%	16.5	tradeable (core)
NAIL	16.7%	0.57	-54.1%	10.9	tradeable (core)
TQQQ	17.7%	0.57	-56.8%	16.4	marginal alternative
FAS	6.8%	0.31	-61.0%	16.5	marginal alternative
SOXL	5.1%	0.27	-71.0%	16.3	marginal alternative
LABU	-2.1%	-0.04	-63.3%	11.1	removed
DUSL	-6.9%	-0.05	-83.7%	9.2	removed

Table 6: Per-ticker CAGR and Sharpe under the SMA + vol gate signal only (no VVIX overlay, no cash routing). LABU and DUSL are removed from the tradeable universe: extended-window Sharpe is negative and, in DUSL’s case, sign inverts between sub-windows (+0.55 on 2021–2026, -0.05 on the extended window).

4.2 Portfolio construction — top-N test

Strategy-return correlations (after signal is applied):

	TECL	NAIL	ROM	TQQQ	FAS	SOXL
TECL	1.00	0.24	0.99	0.82	0.33	0.48
NAIL	0.24	1.00	0.23	0.26	0.28	0.22
ROM	0.99	0.23	1.00	0.82	0.32	0.49

Table 7: TECL and ROM are near-identical strategies (both track XLK). NAIL is the one genuinely uncorrelated instrument.

Top-N portfolios (with VVIX overlay + SGOV cash applied):

Portfolio	CAGR	Sharpe	MaxDD	Notes
TECL+NAIL 50/50	30.9%	1.34	-26.7%	2-ticker baseline
TECL+ROM+NAIL 33/33/33	30.7%	1.34	-26.7%	recommended
Top-4 equal (+ TQQQ)	26.2%	1.20	-28.5%	TQQQ dilutes Tech leg
Top-5 equal (+ SOXL)	22.4%	1.12	-30.4%	further dilution
Top-5 MV-optimal in-sample	26.0%	1.32	-23.6%	concentration risk

Table 8: Adding ROM (2x Tech) as a lower-vol variant of TECL (3x Tech) trades a marginal Sharpe cost for lower sleeve volatility. Top-4 and Top-5 dilute (TQQQ overlaps Tech; SOXL adds correlated risk). MV-optimal weights collapse most weight into ROM but do not survive out-of-sample (see below).

Out-of-sample verification:

OOS test	CAGR	Sharpe	MaxDD
Top-2 MV walk-forward (5y train, 1y test)	29.3%	1.11	-30.7%
Top-3 MV walk-forward	26.9%	1.11	-30.7%
Top-4 MV walk-forward	26.9%	1.11	-30.7%
Top-2 equal-weight, post-2020 (no fitting)	30.9%	1.33	-22.8%
Top-3 equal-weight, post-2020	29.4%	1.34	-18.3%
Top-4 equal-weight, post-2020	27.8%	1.22	-22.4%
Top-5 equal-weight, post-2020	23.9%	1.17	-18.6%

Table 9: MV-optimal weights collapse to Sharpe 1.11 out of sample — in-sample fitting does not generalise. Fixed equal-weight Top-3 wins both in-sample and out-of-sample; out-of-sample drawdown is materially better than Top-2.

4.3 Overlay contributions on the recommended Top-3

Configuration on TECL+ROM+NAIL 33/33/33	CAGR	Sharpe	MaxDD
Baseline (SMA + vol gate only)	34.7%	1.25	-36.4%
+ SGOV 4% cash routing	34.7%	1.25	-36.4%
+ VVIX 3-state overlay	30.7%	1.34	-26.7%
= recommended stack, net of 5bps TC	30.7%	1.34	-26.7%

Table 10: The VVIX overlay is what buys the 10 percentage points of drawdown protection (from -36 to -27) at almost no CAGR cost. SGOV cash routing is a zero-cost yield uplift. Transaction costs at 5 bps per rebalance total 2.4 percentage points over 10.9 years, or roughly 22 basis points per year.

5 VVIX — Why the 3-state design?

VVIX is a leading stress indicator. Historical episodes where the VVIX crossed 100 fired days-to-weeks before realised volatility of XLK / ITB crossed 22%:

- **August 2015** (China devaluation) — VVIX 105 on Aug 21; XLK vol crossed 22% on Aug 24.
- **February 2018** (volmageddon) — VVIX 141 on Feb 5; XLK vol crossed on Feb 6.
- **Q4 2018** (Fed hike scare) — VVIX 115 on Oct 11; realised vol lagged by a week.
- **March 2020** (COVID) — VVIX 200+ multiple days ahead of the realised-vol spike.
- **September 2022** (rate-hike acceleration) — VVIX 130 on Sep 26; realised vol lagged by 3–4 days.
- **April 2025** (tariff shock) — VVIX 108 on Apr 4; realised vol crossed on Apr 8.

5.1 Design bake-off — what we tested

Fifteen VVIX response designs were compared: binary halving, various stepped configurations, sigmoid smooth responses. All net of 5 bps per rebalance, weekly cadence, 5 pp drift band.

Response design	CAGR	Sh	MaxDD	Tx/y	Notes
Binary VVIX > 100 halving (2-state)	30.7%	1.34	-26.7%	8.4	no extreme-VVIX exit
3-state (100 / 130 → 100/50/25)	30.0%	1.32	-26.7%	11.0	still holds at extremes
3-state (100 / 130 → 100/50/0)	29.2%	1.30	-26.7%	11.0	exits too early
3-state (100 / 160 → 100/50/0)	30.7%	1.34	-26.7%	8.8	recommended
3-state (95 / 120 → 100/50/25, tight)	25.5%	1.27	-21.0%	16.3	over-triggered
4-state (100/130/160 → 100/50/25/0)	29.9%	1.32	-26.7%	11.1	no gain over 3-state
Sigmoid centre=100 width=10	22.2%	1.27	-18.1%	21.1	lowest DD but low CAGR
Sigmoid centre=100 width=15	20.2%	1.21	-21.4%	17.6	—

Table 11: VVIX response bake-off, net of transaction costs. The recommended 3-state design (100/160 thresholds) matches binary halving on Sharpe and MaxDD while adding an explicit full-exit state at extreme VVIX. Sigmoid gives lower drawdown but pays 6–8 pp CAGR — it scales down exposure during normal-VVIX periods (80–100) when nothing bad is happening.

5.2 Why exit at 160 and not 130

At the 130 level, VVIX drops back below within a week or two roughly 80% of the time. Most 130 events do not resolve into a full crash. Exiting at 130 costs 0.7 pp of CAGR for no meaningful MaxDD improvement.

At 160, VVIX is at roughly the 99th percentile. Only 18 days in 10.9 years qualify, and every one of them was during a genuine tail event: 2015 China, 2018 volmageddon, 2020 COVID (VVIX peaked at 208), 2022 rate cycle, 2025 tariff shock. Exit here is unambiguous, and the strategy pays essentially nothing to catch these events because they are so rare.

5.3 Why not sigmoid or 4+ states

Sigmoid gives the lowest drawdowns (-18%) but pays 6–8 pp in CAGR because it scales down exposure during normal periods (VVIX 80–100) when there is no follow-through stress. That is over-fitting to drawdown at the cost of return. A 4-state design (100/130/160 → 100/50/25/0) matches the recommended 3-state on all metrics but adds one more trigger boundary — no gain for the extra complexity. Three states (calm / elevated / panic) is the right level of granularity.

6 Independence from the Smart-Beta Book

Under stress, does the leveraged sleeve double up on the smart-beta portfolio’s losses, or decouple? SPY proxies the smart-beta book (long-only US large-cap factor tilts correlate 90–95% with SPY on daily returns). Stress = rolling 21-day SPY drawdown > 10% OR VIX > 30.

Series vs SPY	Overall ρ	Normal ρ	Stress ρ
Raw TECL+ROM+NAIL 33/33/33 buy-and-hold	0.85	0.82	0.92
Strategy-run TECL+ROM+NAIL 33/33/33	0.41	0.51	0.20

Table 12: The SMA + vol gate cuts stress correlation from 0.92 to 0.20 by going to cash exactly when SPY is bleeding. Pre-registered gate 0.60 → PASS.

The signal is doing what it should: during stress, it exits to cash and decouples. The leveraged sleeve is genuine diversification against the smart-beta core.

7 Sizing and Transaction Costs

7.1 Kelly-quarter with 10% NAV cap

Full Kelly ($f^* = \mu/\sigma^2$) for the recommended stack evaluates to $\sim 620\%$ of NAV — absurdly high, as is typical for high-Sharpe backtests. Fractional Kelly is standard practice; the 10% NAV hard cap binds:

$$w = \min(\text{Kelly-quarter}, 0.10 \times \text{NAV}) = 10\% \times \text{NAV}$$

The 10% cap and Kelly-quarter fractional are deliberately more conservative than the Sharpe would justify. Rationale: (a) live variability against backtest is unknown at this point, (b) the operator’s stated posture is a cash-buffered long-only sleeve during the confidence-building phase, (c) the cap can be raised after 12 months of clean live operation.

Because sizing is expressed in % of NAV and the operator invests continuously, this rule works at any NAV size and adjusts automatically as capital accumulates.

7.2 Reduction triggers below 10%

Trigger	Action
Smart-beta regime overlay $\rightarrow$ risk-off	Scale sleeve to 5% NAV
Running sleeve DD $>$ 25% from HWM	Freeze until DD $<$ 15%
Compliance / life event	Manual override at operator’s discretion

Table 13: Triggers layered on top of the VVIX 3-state overlay.

7.3 Transaction cost model

Transaction costs are modelled at **5 basis points per trade**, applied on the notional amount of position change. This covers commission + estimated slippage on limit orders (TECL, ROM, NAIL are all liquid enough to fill at the top of the book; SGOV routing has negligible slippage).

Trigger	Frequency (per year)	Notional change	Cost estimate
Regime state change (ON $\leftrightarrow$ OFF)	8–10	100%	~ 45 bps
VVIX state change (UP $\leftrightarrow$ MID)	1–2	50%	~ 5 bps
VVIX RISK-OFF trigger (VVIX ≥ 160)	0–1	50%	~ 2 bps
Monthly rebalance to 33/33/33	12	$\sim 5\%$ each	~ 8 bps
Drift-band rebalance	1–2	$\sim 5\%$	~ 1 bp
Total estimated transaction cost			~ 60 bps/year

Table 14: Annual transaction cost estimate. Roughly 0.6 percentage points per year, cumulative to ~ 2.7 pp over the 10.9-year backtest. This is included in the headline results.

The cost figure assumes limit orders filled at reference price; a market-order execution style would roughly double the estimate but is not recommended for this sleeve.

8 Live Operating Protocol

Full operational detail — paper trading, phased rollout, alerts, recalibration, interventions — is specified in `docs/ADR-0002-leveraged-standalone-applet.md`. This section summarises the operator’s steady-state workflow.

8.1 Weekly Friday routine (~5 minutes)

1. Applet pulls latest prices at Friday 3:30 PM local; computes regime state and VVIX state.
2. If no state changes since last week: log-only, no action, no alert.
3. If regime state changed (ON $\leftrightarrow$ OFF): Telegram alert; operator opens applet, confirms proposal.
4. If VVIX state changed (UP $\leftrightarrow$ MID, or entry to RISK-OFF): Telegram alert; operator confirms proposal.
5. Any confirmed trade is placed as LMT+GTC on the following Monday morning.

8.2 Telegram alert map

Event	Urgency	Body summary
Regime ON $\rightarrow$ OFF	High	Exit all leveraged to SGOV. Proposal ready.
Regime OFF $\rightarrow$ ON	High	Re-enter to current VVIX state. Proposal ready.
VVIX state RISK_UP $\rightarrow$ RISK_MID	Medium	Halve leveraged exposure. Proposal ready.
VVIX state RISK_MID $\rightarrow$ RISK_OFF (VVIX $\geq$ 160)	Critical	Full exit — deep panic event. Proposal ready.
VVIX state RISK_OFF $\rightarrow$ RISK_MID (VVIX $<$ 160)	High	Re-enter to RISK_MID (50% cap). Proposal ready.
VVIX state RISK_DOWN $\rightarrow$ RISK_MID	Medium	Restore to 50%.
VVIX state RISK_MID $\rightarrow$ RISK_UP	Medium	Restore to full sleeve cap.
Monthly rebalance due	Low	Restore 33/33/33 within sleeve.
Running sleeve DD $>$ 20%	High	DD approaching freeze trigger.
Running sleeve DD $>$ 25%	Critical	Freeze triggered. Sleeve suspended.
Order rejection or unknown status	High	Manual investigation required.

Table 15: Alerts drive weekly operator attention. Steady-state volume is 1–5 alerts per week (many ”state OK” heartbeats during calibration); genuine stress events produce 5–15 alerts in a short burst.

9 Limitations and Future Work

9.1 Limitations

- Portfolio backtest bounded by NAIL inception (2015-08). TECL and ROM alone have 16.5 years.
- Two-underlying exposure (Tech via XLK, Homebuilders via ITB). The Tech leg uses two vehicles (TECL 3x + ROM 2x) but one underlying. Further diversification must come from outside the leveraged sleeve.
- SGOV yield modelled as constant 4%; live yield varies. Effect on CAGR degrades from ~ 2 pp to ~ 1 pp if yield falls below 2%.
- VVIX overlay uses close-to-close prices; intra-day tick spikes above 100 do not trigger — the trigger is the Friday close.
- Correlation with smart-beta uses SPY as a proxy; a full smart-beta simulator would sharpen but not change the sign of the result.

9.2 Future work

- **Standalone applet build.** ADR-0002 spec ready to implement (Streamlit surface, paper trading gate, recalibration cycle, alerts, interventions).
- **Parallel non-equity sleeves.** TMF (3x TLT bonds), EDC (3x EM), JNUG (gold miners) could carry the same signal template calibrated for their vol regimes. Each is a fresh calibration exercise — deferred to a future ADR.

- **Live tracking vs backtest.** After 12 months live, compare live Sharpe / CAGR / MaxDD to backtest expectation; freeze and review if any diverges by more than the pre-registered band.
- **Synthetic pre-inception NAIL returns** from $ITB \times 3$ with decay adjustment; would extend backtest depth by 5 years.

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